

How Much of Multi-Lead Fetal QRS Detection Does One Lead Recover? A Controlled Audit on 22 Women and 60 Cross-Recorder Records

Ngo Binh Minh¹, [Co-author]¹, [Supervisor]¹

¹Faculty of Information Technology, Ton Duc Thang University, Ho Chi Minh City, Vietnam

Abstract

Multi-channel source separation is the usual answer to fetal QRS (fQRS) detection from abdominal ECG; a one-lead device cannot use it. We ask how much of that benefit a small single-lead network recovers, and measure it like-for-like: the published Power-MF pipeline is re-run by us under GNU Octave on 22 women (ADFECGDB and the Silesian database, duplicates removed) and scored with the same ± 50 ms matcher as our 113 481-parameter dilated temporal convolutional network. Restricting Power-MF to one lead costs it 12.12 F1 points (98.83 to 86.71); the network, on one label-blind lead, scores 97.56 and recovers 10.85 of those points, 89.5% (95% CI 81.4–103.2), so it is not separable from four-channel Power-MF (-1.27 , CI $[-3.08; +0.27]$, $p = 0.16$, better in 18 of 22 women, median $+0.23$) and ahead of one-channel Power-MF ($+10.85$, 22 of 22). Across recorders the picture changes. CinC 2013 set-a is not independent of ADFECGDB, as its organisers documented; we identify by measurement the 15 records that are verbatim copies of our training data and score the other 60: training on 22 women instead of five raises the label-blind score from 64.04 to 74.28 ($p = 3.5 \cdot 10^{-10}$), the oracle lead reaches 83.60, and the lead rule fails before the detector does. A lead rule pre-specified as confirmatory gains $+6.44$ but misses Holm correction ($p_{\text{Holm}} = 0.051$); a post-hoc rule driven by the detector's own peak probabilities gains $+7.73$ ($p_{\text{Holm}} = 0.0039$) and is reported as a hypothesis. At matched parameter count, two receptive-field variants are equivalent to the network and a non-dilated CNN is worse only on the logit scale; four unsupervised adaptation methods recover none of the cross-recorder gap.

1. Introduction

Fetal heart rate from abdominal electrodes needs a reliable fQRS detector, and a single-lead one keeps the hardware small. The PhysioNet/CinC Challenge 2013 fixed the metric still in use, F1 with a ± 50 ms window [1, 2]; Behar et al. benchmarked template subtraction, Kalman and blind-source methods under it [3], and Andreotti et al. showed that pre-processing changes the ranking of extraction algorithms [4]. Deep single-lead detectors since [5] include dual-path source separation [6] and guided empirical mode decomposition [7]; the strongest recent multi-channel pipeline, Power-MF [8], combines two ICA stages with a matched filter and a power-spectral-density (PSD) lead rule. Its numbers, like most, are quoted across papers rather than measured under one matcher, and the lead is often chosen with the test labels.

The ingredients of our detector are established elsewhere and we claim none as new: per-sample R-peak output [9], dilation as the useful architectural knob [10], the 10–60 Hz band [7]. What we contribute is measurement:

- the fraction of the multi-channel benefit one lead recovers, with the competitor executed by us on the same recordings and scored with the same matcher, per woman;
- a cross-recorder evaluation on the 60 CinC 2013 set-a records that are not copies of the training data, separating detector loss from lead-rule loss, with the confirmatory and the post-hoc lead rule both reported;
- matched-parameter architecture comparisons on the logit scale, and a negative result for four unsupervised domain-adaptation methods.

2. Methods

2.1. Data

ADFECGDB [11]: five labour recordings (r01, r04, r07, r08, r10), four abdominal leads at 1 kHz, 5 min each, fQRS annotated from a scalp electrode. **Silesian database** [12]:

ten pregnancy recordings (B1, 32–42 weeks, 199.6 min) and twelve labour recordings (B2, 5 min), four abdominal leads at 500 Hz, same hospital and recorder as ADFECGDB. B1 annotations are indirect (author pipeline plus expert correction) and in 5/10 records sit 8–12 ms after the abdominal R peak; B2 annotations come from a scalp electrode. The overlap with ADFECGDB was noted by Baldazzi and Pani [13]; cross-correlating all 4×4 lead pairs we find that five B2 records are ADFECGDB records (B2_01 = r01, B2_02 = r10, B2_07 = r04, B2_10 = r07, B2_11 = r08; correlation 0.988–0.994). The independent set is therefore 22 women (5 ADFECGDB, 10 B1, 7 new B2).

CinC 2013 set-a [1, 2]: 75 records, four channels at 1 kHz, 1 min each, a different recorder. Set-a is *not* independent of ADFECGDB, and this was documented by the Challenge organisers: 25 of the 447 Challenge records came from the “Abdominal and Direct FECG” collection (Table 1 of [1], Table 2 of [2]), and Clifford et al. cautioned that an entry trained on it “may have led to a bias in the results as this database was included in set-a, set-b (and possibly a few records in set-c)” [2]. No accessible source states *which* set-a records they are. By normalised cross-correlation of the raw signals over all lags, in two independent implementations, exactly 15 — a03–a05, a08, a12–a15, a17, a19, a20, a22–a25 — are verbatim copies (NCC = 1.0000 on all four channels, $|\Delta\text{RR}| = 0.0$ ms) of the 0–60, 120–180 and 240–300 s windows of r01, r04, r07, r08 and r10; the other 60 reach at most 0.62. Those 15 records score 99.87 under our 22-woman model and would have inflated 75-record means by 3.27–7.18 points. Every CinC figure below uses the 60 remaining records; a 53-record variant also removes a33, a38, a47, a52, a54, a71 and a74, whose reference annotations are known to be unreliable [3] (list fixed before the run).

2.2. Detector

One abdominal lead is band-passed 10–60 Hz (4th-order Butterworth, zero-phase) with a 50 Hz notch and resampled

to 250 Hz. Maternal QRS are located on an 8–25 Hz envelope and removed by subtracting a median beat template with per-beat least-squares scaling. The residual and the filtered signal, each robust-scaled, form a two-channel 4 s input. The network is a dilated residual TCN: a stem and five residual blocks (dilations 1, 2, 4, 8, 16; 40 channels; kernel 7; BatchNorm; GELU) with a 1×1 output convolution, 113 481 parameters, receptive field 1 516 ms. It emits one logit per sample against a Gaussian heat-map target ($\sigma = 12$ ms) at each annotated fQRS [9], trained with weighted binary cross-entropy and AdamW (one-cycle). At inference, overlapping windows are averaged and peaks above a threshold with a 250 ms refractory period are the fQRS (0.48 MB, 4.35 ms per window on a desktop CPU).

2.3. Protocol and statistics

Two models are used. The *five-woman model* (m5) is leave-one-record-out (LORO) on ADFECGDB: one test, one inner validation and three training records, six epochs; Silesian and CinC records are scored zero-shot. The *22-woman model* (m22) is trained on all 22 independent women with 11-fold grouped cross-validation (19 train, 1 validation, 2 test per fold, four epochs, on-the-fly augmentation); CinC is scored zero-shot by the production checkpoint. In both, the threshold is chosen on the validation subject from a grid and applied unchanged. Detections are matched one-to-one to annotations by greedy nearest matching within ± 50 ms. Because a device sees one lead, the headline lead is *label-blind*: following Power-MF [8], the lead whose residual envelope has the largest Welch PSD peak in 1.8–3.0 Hz; the oracle lead (best F1 per record, uses labels) is shown only as a ceiling.

The unit of analysis is the woman (the record for CinC), never the (record $\times$ lead) pair. We report mean differences with 95% cluster-bootstrap percentile intervals (10 000 resamples, seed 0) and exact paired Wilcoxon tests; multiple lead rules are Holm-corrected over the family. “No difference” claims use two one-sided tests (TOST) at a 1.0-point margin. Because F1 saturates near 100, architecture comparisons are repeated on the logit scale with a continuity correction of $1/(4n_{\text{ref}})$. With $n = 5$ the smallest two-sided exact signed-rank p is 0.0625, so no ADFECGDB comparison can reach $p < 0.05$; we quote intervals there.

2.4. Experiments

Power-MF [8] was not quoted but executed: the authors’ MATLAB pipeline was run under GNU Octave 11.3 and scored with *our* matcher, at four leads with both ICA stages (PMF-4) and restricted to the single PSD-selected lead (PMF-1). Octave’s `findpeaks` builds an $O(k^2)$ distance matrix for `MinPeakDistance` and overflows on long B1 recordings; we re-implemented the MATLAB semantics in $O(k)$ (48/48 test cases). The port scores 99.40 ± 0.51 on the ten B1 records against 99.46 published by the authors. The *recovery ratio* is (ours – PMF1)/(PMF4 – PMF1), bootstrapped over women.

Lead rules. Seven label-blind alternatives to the PSD rule were fixed in a written plan before any was evaluated on CinC, together with the decision rule “confirmatory rule = the one with the highest in-domain F1 over 22 women”. The plan was written after per-lead F1 had been computed and is not anchored by a third party, so we call it a written plan and nothing stronger. Rules read only the detector’s output and classical signal-quality indices (SQIs): *gate* (a reject

Table 1: Macro F1 (%), ± 50 ms, label-blind PSD lead and mean over four leads. m5: trained on five ADFECGDB women (LORO in domain, zero-shot elsewhere); m22: 11-fold grouped CV over 22 women, zero-shot on CinC. CI: 95% bootstrap over subjects (records for CinC), m5 PSD lead.

Configuration	n	m5		m22		m5 CI
		PSD	4-lead	PSD	4-lead	
ADFECDGB labour	5	99.21	97.45	99.40	98.69	—
Silesia B2 labour, 7 new women	7	95.87	93.73	96.82	94.50	[88.4; 99.8]
Silesia B1 pregnancy	10	93.30	91.31	97.15	97.01	[84.9; 99.4]
All 22 women	22	95.46	93.47	97.56	96.59	—
CinC 2013 set-a, 60 clean records	60	64.04	56.00	74.28	67.69	[55.4; 72.5]

gate trained on PSD-lead segments; keep the lead with the highest $1 - \bar{P}(\text{bad})$), *gate4* (same, trained on all four leads), *rrcv* (lowest RR-interval coefficient of variation), *peakprob* (highest mean detector probability at its own detected peaks; no training, no hyper-parameter), *rrplaus*, a learned selector and a four-lead fusion.

Architecture. Six alternatives to the TCN at matched parameter count ($\pm 2.7\%$) on the 22 women: a wide (3 052 ms) and a narrow (748 ms) receptive field, a three-branch multi-scale TCN, a non-dilated CNN widened to the same 1 508 ms field, a 1-D U-Net and a small-field CNN (60 ms). Three folds, three epochs, one seed, so the run is a scaled-down replicate of the production protocol. The band was also varied on the TCN (10–60 vs 1–45 Hz, four epochs).

Domain adaptation, hyper-parameters chosen on a 12-woman-to-B1 transfer and never on CinC: an adaptive mains notch (50 and/or 60 Hz, whichever the record’s spectrum shows), confidence-thresholded self-training, AdaBN [14] and TENT [15].

Reject gate. A 4 s segment is *bad* if the detector’s F1 on it is below 80. A gradient-boosted classifier on 12 classical SQIs [16] is trained leave-one-subject-out over the 22 women (3 890 segments, 5.40% bad) and scored both pooled and within record; intervals resample subjects.

3. Results

3.1. Detection across data configurations

Table 1 gives both models on every configuration. In domain the label-blind PSD rule sits within 0.02–1.90 points of the oracle lead per set (1.08 pooled over 22 women). Zero-shot from five labour women to ten pregnancy women of the same recorder costs about six points (99.21 to 93.30); training on 22 women closes most of it (97.15). A second seed of m22 moves the 22-woman mean from 97.56 to 97.59 (+0.03, CI [−0.18; 0.34]; largest per-woman change 2.81, B2_03).

3.2. How much of the multi-channel benefit one lead recovers

Table 2 is the like-for-like comparison. Restricting Power-MF to one lead costs it 12.12 F1 points (98.83 to 86.71, CI [−18.27; −6.90], worse in 21 of 22 women): that is the measured price of the four-channel ICA stages on this material. Our single-lead network scores 97.56 and recovers 10.85 of those points (+10.85 over PMF-1, CI [+6.80; +15.40], $p = 4.8 \cdot 10^{-7}$, 22 of 22 women), a recovery ratio of 89.5% (cluster-bootstrap CI 81.4–103.2; leave-one-woman-out range 88.6–93.4). Against the full four-channel pipeline it is not separable (−1.27, CI [−3.08; +0.27], $p = 0.16$); on ADFECGDB the interval excludes zero in its favour (+0.39, CI [+0.22; +0.60]),

Table 2: Power-MF [8] *re-run* by us at four channels (PMF-4) and restricted to one lead (PMF-1) against the 22-woman single-lead model: same recordings, same ± 50 ms matcher. Δ_4 , Δ_1 : ours – Power-MF per woman, 95% cluster-bootstrap CI (records for CinC). Power-MF was not run at four channels on CinC.

Subject set	n	mean F1 (%)			ours – Power-MF	
		PMF-4	PMF-1	ours	Δ_4 [95% CI]	Δ_1 [95% CI]
ADFECDGB	5	99.01	92.37	99.40	+0.39 [+0.22; +0.60]	+7.02 [+3.84; +9.34]
Silesia B2, new	7	97.90	87.28	96.82	-1.08 [-4.04; +0.54]	+9.54 [+3.09; +17.9]
Silesia B1	10	99.40	83.48	97.15	-2.25 [-5.55; +0.32]	+13.68 [+6.91; +21.3]
All 22	22	98.83	86.71	97.56	-1.27 [-3.08; +0.27]	+10.85 [+6.80; +15.4]
CinC 2013, 60 clean records	60	—	55.97	74.28	—	+18.32 [+13.4; +23.6]

5/5).

The mean deficit is not uniform: the median difference over 22 women is +0.23 and the network is better in 18. Three recordings — B1_07 (-12.88), B1_06 (-10.51), B2_03 (-9.77) — pull the mean below zero. The defensible statement is that one lead recovers most of the multi-channel benefit, not that it matches four leads on every woman.

3.3. CinC 2013: the 60 records that are not training data

Under the label-blind PSD rule, training on 22 women instead of five raises the mean from 64.04 to 74.28 (+10.24, CI [7.13; 13.60], $p = 3.5 \cdot 10^{-10}$, better in 52 records, worse in 3); records at or above 90 go from 27 to 33 and those below 50 from 22 to 16; the 53-record variant gives 64.19 to 75.27 (+11.08, $p = 4.4 \cdot 10^{-9}$). The median record of m22 is 95.07: the distribution is split, not shifted. One-lead Power-MF on the same 60 records scores 55.97 (Table 2).

The lead rule loses accuracy across recorders before the detector does: the PSD rule is 9.32 points below the oracle lead (74.28 vs 83.60, CI [5.10; 14.15]) and picks it in only 17 of 60 records. Table 3 shows four of the seven pre-specified rules; the three omitted (rrplaus, learned, fuse) are the weakest and are in the released JSON. The written decision rule selected *gate* as confirmatory; it gains +6.44 over PSD (CI [+2.49; +11.10], $p = 0.010$) but *does not survive* Holm correction over the seven rules ($p_{\text{Holm}} = 0.051$). We report this as a failed confirmatory test. Within the same family, *gate4* (81.01, $p_{\text{Holm}} = 0.015$) does survive: the family-level test succeeds while the single designated rule does not. The in-domain arbiter was also uninformative by construction: the four best rules lie within 0.04 points of each other on 22 women, where the whole oracle headroom is 1.08 points, a design error on our part. The strongest of the seven is *peakprob*: 82.01 against 74.28, +7.73 (CI [+3.82; +12.41], $p = 5.6 \cdot 10^{-4}$, $p_{\text{Holm}} = 0.0039$ over the family), 82.9% of the oracle headroom, records below 50 from 16 to 9, 19 wins / 35 ties / 6 losses, no record losing more than 3.90 points. *peakprob* was chosen after the CinC results were seen; the reason to still report it is that the effect survives Holm correction over the whole pre-specified family. On the logit scale over the 22 in-domain women the ranking changes — *gate* leads under two of three record pairings and *peakprob* is third (+0.13 logit vs PSD, CI [-0.04; +0.41]) — so the post-hoc problem does not disappear. No public dataset with real fQRS labels exists on which to replicate it (NIFEADB, NInFEA, nifecgdb and set-b ship none), so it remains a strong, unconfirmed hypothesis.

Table 3: CinC 2013 set-a, 60 records that are not ADFECGDB copies, zero-shot. Δ : per-record difference against the PSD rule under m22, 95% bootstrap over records, exact signed-rank p Holm-corrected over the seven pre-specified rules; W/T/L wins, ties, losses; “ ≥ 90 ”/“ < 50 ” count m22 records. All rules are label-blind except the oracle.

Lead rule	status	m5	m22	Δ vs PSD [95% CI]	p_{Holm}	W/T/L	≥ 90 / < 50
PSD [8]	headline	64.04	74.28	—	—	—	33 / 16
4-lead mean	reference	56.00	67.69	-6.59	—	—	15 / 17
<i>gate</i>	confirmatory, failed	—	80.72	+6.44 [+2.49; +11.10]	0.051	16/37/7	38 / 12
<i>gate4</i>	pre-specified	—	81.01	+6.72 [+2.85; +11.18]	0.015	17/38/5	38 / 12
<i>rrcv</i>	pre-specified	—	80.00	+5.72 [+1.61; +10.41]	0.41	15/35/10	38 / 12
<i>peakprob</i>	post-hoc choice	—	82.01	+7.73 [+3.82; +12.41]	0.0039	19/35/6	38 / 9
Oracle lead	uses labels	72.76	83.60	+9.32 [+5.10; +14.15]	—	25/35/0	38 / 8
PSD, 53 records	declared variant	64.19	75.27	—	—	—	31 / 14

Table 4: Six architectures against the TCN at matched parameter count, 22 women, PSD lead, three folds, three epochs, one seed. Δ F1 and Δ logit are alternative – TCN per woman with 95% cluster-bootstrap CI; TOST at ± 1.0 F1, Holm over six.

Model	params	RF (ms)	F1	Δ F1 [95% CI]	Δ logit [95% CI]	W/T/L	TOST
TCN (ours)	113 481	1 516	97.64	—	—	—	—
TCN, wide field	116 514	3 052	97.63	-0.01 [-0.18; +0.16]	+0.09 [-0.18; +0.29]	11/5/6	equivalent
TCN, multi-scale	116 011	2 716	97.62	-0.02 [-0.34; +0.24]	+0.04 [-0.25; +0.27]	9/8/5	equivalent
TCN, narrow field	114 886	748	96.93	-0.71 [-1.73; +0.07]	-0.29 [-0.68; +0.12]	4/4/14	inconclusive
CNN, no dilation, wide	114 943	1 508	96.84	-0.80 [-2.40; +0.13]	-0.45 [-0.75; -0.16]	6/1/15	inconclusive
U-Net 1-D	114 247	636	96.38	-1.25 [-2.90; -0.19]	-0.80 [-1.17; -0.43]	3/1/18	inconclusive
CNN, small field	114 493	60	94.53	-3.10 [-6.19; -0.96]	-1.64 [-2.09; -1.16]	2/0/20	inconclusive

3.4. Architecture at matched parameters, on the logit scale

Table 4 compares six alternatives to the TCN on the 22 women. Two are *equivalent* under TOST after Holm correction: the wide receptive field (3 052 ms, -0.01 F1, $p_{\text{Holm}} 7 \cdot 10^{-10}$) and the multi-scale TCN (-0.02, $4.6 \cdot 10^{-6}$); beyond 1.5 s more context buys nothing. Two are worse on both scales: the 1-D U-Net (-1.25, logit -0.80 [-1.17; -0.43]) and the 60 ms CNN (-3.10, worse in 20 of 22, logit -1.64). The instructive case is the non-dilated CNN widened to the same 1 508 ms field: on the F1 scale it ties (-0.80, CI [-2.40; +0.13], $p_{\text{Holm}} = 0.30$), on the logit scale it is worse (-0.45, CI [-0.75; -0.16]), and the narrow field (748 ms) sits between (-0.71; logit -0.29 [-0.68; +0.12]). F1 near 100 hides differences that the logit scale resolves, so every architecture claim here is stated on both. Changing the band on the TCN from 1–45 to 10–60 Hz gives +2.44 on the PSD lead (CI [-0.04; 6.29]; three women carry all of it) and -0.07 [-0.51; 0.39] averaged over four leads: the band is not an established effect on this network.

3.5. Domain adaptation, reject gate, clinical read-out

On the 60 CinC records none of the four unsupervised methods improves the PSD-lead score of 74.28: adaptive notch +0.25 (CI [-0.25; +1.08], $p = 0.70$), self-training -0.67 [-1.11; -0.29], AdaBN -1.58 [-2.50; -0.73], TENT -2.43 [-3.64; -1.35]. The three measured shifts between the datasets (60 s duration, 10.13-bit quantisation, 60 Hz mains at 1.64 times the noise floor), imposed on the in-domain recordings, cost 0.01 points. Whether the remaining gap is fetal signal absent from the lead or signal the model misses could not be decided: the visibility test built for that purpose has an 18.0% false-negative rate (CI 12.1–25.0) on these records.

The LOSO reject gate reaches a pooled AUROC of 0.965 (CI [0.857; 0.992]) and a within-record AUROC of 0.934 (CI [0.872; 0.981], over the 11 women with both classes). Ranked by gate score, the three women that carry the deficit against four-channel Power-MF (B2_03, B1_07, B1_06) are the three least reliable of 22. Short-term variability computed from

detected beats is biased +0.33 ms on ADFECGDB (limits of agreement $[-0.89; +1.55]$); on an earlier ten-record CinC subsample the bias was +20.50 ms, so the detector is usable for trending on gated records and not as a standalone STV monitor.

4. Discussion

One lead recovers most of what four leads buy. Multi-channel separation is worth a directly measured 12.12 F1 points to Power-MF itself; a 113 k single-lead network recovers 10.85 of them. The residual gap is concentrated in three of 22 recordings, and the reject gate ranks exactly those three as least reliable, which is what a one-lead device must do: fail loudly, not often.

Across recorders the lead rule fails before the detector. Training on 22 women recovers 10.24 points on CinC without closing the recorder gap; the PSD rule is within 1.90 points of the oracle in domain and 9.32 below it out of domain. A rule that lets the detector score the leads is the obvious repair, but the confirmatory version of that idea missed Holm correction and the version that passed was chosen after the data were seen; the next held-out set must be opened with the rule fixed beforehand. Four adaptation methods that repair the shifts we can measure recover nothing, and the shifts themselves cost 0.01 points in domain; we do not conclude from this that the signal is missing, because the test that would show it has an 18% false-negative rate.

Architecture is a small variable, and F1 is the wrong scale to see it. At matched parameters two variants are equivalent to the TCN and one (dilation at equal field) differs only on the logit scale. Nothing here is claimed as new; the receptive field near 1.5 s and dilation are the two design choices we can resolve with 22 women.

Limitations. (i) n : five women on ADFECGDB and 22 in all, one hospital, one recorder; every interval is wide and no ADFECGDB comparison can reach $p < 0.05$ at subject level. (ii) Against four-channel Power-MF the point estimate is -1.27 ; the interval contains zero, but the sign is against us. Power-MF ran under Octave at our resampled rate with our matcher, not in the authors' evaluation. (iii) The lead-rule plan was written after per-lead F1 existed and is not third-party anchored; *peakprob* is post hoc; no third labelled dataset exists to confirm it. (iv) The architecture and band runs use three folds, three to four epochs and one seed; the main model one seed (a second seed moved the mean by 0.03). (v) B1 labels are indirect, so B1 F1 measures agreement with the authors' pipeline. (vi) Whether the 60 clean CinC records come from 60 distinct women was not checked; $NCC \leq 0.62$ excludes verbatim copies only. (vii) The reject gate has not been re-scored on the 60 clean CinC records; gate figures computed earlier on the contaminated 75-record set are withdrawn with the rest.

5. Conclusion

Measured under one matcher on 22 women, four-channel Power-MF gains 12.12 F1 points over its own single-lead restriction and a 113 k single-lead network recovers 10.85 of them (89.5%): not separable from four channels (-1.27 , CI $[-3.08; +0.27]$), ahead of one channel in 22 of 22 women. On the 60 CinC 2013 set-a records that are not copies of the training data the label-blind score is 74.28 after training on

22 women, 9.32 below the oracle lead: cross-recorder lead selection, not the detector, is the open problem, and the best candidate rule is a hypothesis awaiting a labelled third dataset.

References

- [1] Silva I, Behar J, Sameni R, Zhu T, Oster J, Clifford GD, et al. Noninvasive fetal ECG: the PhysioNet/Computing in Cardiology Challenge 2013. In: Computing in Cardiology, vol. 40; 2013. p. 149-52. ISSN 2325-8861.
- [2] Clifford GD, Silva I, Behar J, Moody GB. Non-invasive fetal ECG analysis. Physiological Measurement. 2014;35(8):1521-36.
- [3] Behar J, Oster J, Clifford GD. Combining and benchmarking methods of foetal ECG extraction without maternal or scalp electrode data. Physiological Measurement. 2014;35(8):1569-89.
- [4] Andreotti F, Behar J, Zaunseder S, Oster J, Clifford GD. An open-source framework for stress-testing non-invasive foetal ECG extraction algorithms. Physiological Measurement. 2016;37(5):627-48.
- [5] Zhong W, Liao L, Guo X, Wang G. A deep learning approach for fetal QRS complex detection. Physiological Measurement. 2018;39(4):045004.
- [6] Shokouhmand A, Tavassolian N. Fetal electrocardiogram extraction using dual-path source separation of single-channel non-invasive abdominal recordings. IEEE Transactions on Biomedical Engineering. 2023;70(1):283-95.
- [7] Xu X, Zhang Y, Rao Y, Xu Y, Gao Y, Tu H. A deep learning-guided ensemble empirical mode decomposition method for single-channel fetal electrocardiogram extraction. Sensors. 2026;26(7):2037.
- [8] Jaeger KM, Nissen M, Rahm S, Titzmann A, Fasching PA, Beilner J, et al. Power-MF: robust fetal QRS detection from non-invasive fetal electrocardiogram recordings. Physiological Measurement. 2024;45(5):055009.
- [9] Zahid, Kiranyaz, Ince, et al. Robust R-peak detection in low-quality holter ECGs using 1D convolutional neural network. IEEE Transactions on Biomedical Engineering. 2022;69(1):119-28.
- [10] Fotiadou E, van Sloun RJG, van Laar JOEH, Vullings R. A dilated inception CNN-LSTM network for fetal heart rate estimation. Physiological Measurement. 2021;42(4):045007.
- [11] Goldberger AL, Amaral LAN, Glass L, Hausdorff JM, Ivanov PC, Mark RG, et al. PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals. Circulation. 2000;101(23):e215-20.
- [12] Matonia A, Jezewski J, Kupka T, Jezewski M, Horoba K, Wrobel J, et al. Fetal electrocardiograms, direct and abdominal with reference heartbeat annotations. Scientific Data. 2020;7:200.
- [13] Baldazzi G, Pani D. Open data: valuable resources and opportunities for the researchers in fetal cardiac monitoring. In: Innovative Technologies and Signal Processing in Perinatal Medicine. Springer; 2023.
- [14] Li Y, Wang N, Shi J, Hou X, Liu J. Adaptive batch normalization for practical domain adaptation. Pattern Recognition. 2018;80:109-17.
- [15] Wang D, Shelhamer E, Liu S, Olshausen B, Darrell T. Tent: fully test-time adaptation by entropy minimization. In: International Conference on Learning Representations (ICLR); 2021. ArXiv:2006.10726.
- [16] Andreotti, Grässer, Malberg, Zaunseder. Non-invasive fetal ECG signal quality assessment for multichannel heart rate estimation. IEEE Transactions on Biomedical Engineering. 2017.